

DBMS QUESTIONS

Name : Harsh

Roll No : 2501730062

Course : B.Tech CSE AIML

Section : B

Sem : 3 (III)

Q1. Display student names along with their department names.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Schemas

- Filter objects
- college
- college_management
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1 x student

```
1 SELECT
2     Student.student_name,
3     Department.department_name
4 FROM Student
5 INNER JOIN Department
6 ON Student.department_id = Department.department_id;
```

Result Grid

	student_name	department_name
▶	Aman	Computer Science
	Rahul	Computer Science
	Priya	Computer Science
	Neha	Computer Science
	Arjun	Computer Science
	Karan	Computer Science
	Simran	Information Technology
	Riya	Information Technology
	Pooja	Information Technology
	Vikas	Electronics
	Ankit	Electronics

Administration Schemas

Information: No object selected

Result 3 x Read Only Context Help Snippets

Output: Action Output

#	Time	Action	Message	Duration / Fetch
✓ 10	19:51:39	INSERT INTO Course VALUES (201, 'Database Management Systems', ...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.000 sec
✓ 11	19:51:51	INSERT INTO Enrollment VALUES (301, 101, 201, 85), (302, 101, 202, 7...	18 row(s) affected Records: 18 Duplicates: 0 Warnings: 0	0.016 sec
✓ 12	19:52:02	USE college_management	0 row(s) affected	0.000 sec
✓ 13	19:52:02	SELECT * FROM Department LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
✓ 14	19:52:12	SELECT * FROM Student LIMIT 0, 1000	13 row(s) returned	0.000 sec / 0.000 sec
✓ 15	19:52:46	SELECT Student.student_name, Department.department_name FR...	11 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQLAdditions: Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid
Form Editor
Field Types
Query Stats

Q2. Display all students and their corresponding department locations.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: SCHEMAS

- college
- college_management
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1 student

```
1 • SELECT
2   Execute the selected portion of the script or everything, if there is no selection
3   Department.location
4 FROM Student
5 INNER JOIN Department
6 ON Student.department_id = Department.department_id;
```

Result Grid

	student_name	location
▶	Aman	Delhi
	Rahul	Delhi
	Priya	Delhi
	Neha	Delhi
	Arjun	Delhi
	Karan	Delhi
	Simran	Noida
	Riya	Noida
	Pooja	Noida
	Vikas	Gurgaon
	Ankit	Gurgaon

Administration Schemas

Information

No object selected

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 11	19:51:51	INSERT INTO Enrollment VALUES (301, 101, 201, 85), (302, 101, 202, 7...	18 row(s) affected Records: 18 Duplicates: 0 Warnings: 0	0.016 sec
✓ 12	19:52:02	USE college_management	0 row(s) affected	0.000 sec
✓ 13	19:52:02	SELECT * FROM Department LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
✓ 14	19:52:12	SELECT * FROM Student LIMIT 0, 1000	13 row(s) returned	0.000 sec / 0.000 sec
✓ 15	19:52:46	SELECT Student.student_name, Department.department_name FR...	11 row(s) returned	0.000 sec / 0.000 sec
✓ 16	19:54:29	SELECT Student.student_name, Department.location FROM Stude...	11 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Q3. Display the details of all courses along with their department names.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1 x student

Limit to 1000 rows

```
1 • SELECT
2   Course.course_id,
3   Course.course_name,
4   Department.department_name
5 FROM Course
6 INNER JOIN Department
7 ON Course.department_id = Department.department_id;
```

Result Grid

	course_id	course_name	department_name
▶	201	Database Management Systems	Computer Science
	202	Data Structures	Computer Science
	203	Operating Systems	Computer Science
	204	Web Development	Information Technology
	205	Computer Networks	Information Technology
	206	Digital Electronics	Electronics
	207	Thermodynamics	Mechanical

Administration Schemas

Information

No object selected

Result 5 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	19:52:02	USE college_management	0 row(s) affected	0.000 sec
13	19:52:02	SELECT * FROM Department LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
14	19:52:12	SELECT * FROM Student LIMIT 0, 1000	13 row(s) returned	0.000 sec / 0.000 sec
15	19:52:46	SELECT Student.student_name, Department.department_name FR...	11 row(s) returned	0.000 sec / 0.000 sec
16	19:54:29	SELECT Student.student_name, Department.location FROM Stude...	11 row(s) returned	0.000 sec / 0.000 sec
17	19:55:09	SELECT Course.course_id, Course.course_name, Department.d...	7 row(s) returned	0.000 sec / 0.000 sec

Q4. Display student name, course name, and marks obtained by each student.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: SCHEMAS

- Filter objects
- college
- college_management
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1 x student

```
1 • SELECT
2     Student.student_name,
3     Course.course_name,
4     Enrollment.marks
5 FROM Student
6 INNER JOIN Enrollment
7     ON Student.student_id = Enrollment.student_id
8 INNER JOIN Course
9     ON Enrollment.course_id = Course.course_id;
```

Result Grid

student_name	course_name	marks
Aman	Database Management Systems	85
Rahul	Database Management Systems	92
Priya	Database Management Systems	76
Neha	Database Management Systems	65
Karan	Database Management Systems	55
Aman	Data Structures	78
Rahul	Data Structures	88
Arjun	Data Structures	95
Aman	Operating Systems	91
Priya	Operating Systems	84
Arjun	Operating Systems	89
Neha	Web Development	72
Simran	Web Development	81
Riya	Web Development	68

Result 6 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 13	19:52:02	SELECT * FROM Department LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
✓ 14	19:52:12	SELECT * FROM Student LIMIT 0, 1000	13 row(s) returned	0.000 sec / 0.000 sec
✓ 15	19:52:46	SELECT Student.student_name, Department.department_name FR...	11 row(s) returned	0.000 sec / 0.000 sec
✓ 16	19:54:29	SELECT Student.student_name, Department.location FROM Stude...	11 row(s) returned	0.000 sec / 0.000 sec
✓ 17	19:55:09	SELECT Course.course_id, Course.course_name, Department.d...	7 row(s) returned	0.000 sec / 0.000 sec
✓ 18	19:55:40	SELECT Student.student_name, Course.course_name, Enrollme...	18 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQL Additions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Q5. Display all students who are enrolled in courses.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: SCHEMAS

- Filter objects
- college
- college_management
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1 student

```
1 • SELECT DISTINCT
2     Student.student_name
3 FROM Student
4 INNER JOIN Enrollment
5     ON Student.student_id = Enrollment.student_id;
```

Result Grid

student_name
Aman
Rahul
Priya
Neha
Arjun
Karan
Simran
Riya
Vikas
Ankit
Pooja

Administration Schemas

Information: No object selected

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 14	19:52:12	SELECT * FROM Student LIMIT 0, 1000	13 row(s) returned	0.000 sec / 0.000 sec
✓ 15	19:52:46	SELECT Student.student_name, Department.department_name FR...	11 row(s) returned	0.000 sec / 0.000 sec
✓ 16	19:54:29	SELECT Student.student_name, Department.location FROM Stude...	11 row(s) returned	0.000 sec / 0.000 sec
✓ 17	19:55:09	SELECT Course.course_id, Course.course_name, Department.d...	7 row(s) returned	0.000 sec / 0.000 sec
✓ 18	19:55:40	SELECT Student.student_name, Course.course_name, Enrollme...	18 row(s) returned	0.000 sec / 0.000 sec
✓ 19	19:56:19	SELECT DISTINCT Student.student_name FROM Student INNER JOI...	11 row(s) returned	0.015 sec / 0.000 sec

Object Info Session

SQLAdditions: Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid Form Editor Field Types Query Stats

Read Only Context Help Snippets

INNER JOIN Questions

Q6. Use INNER JOIN to display:

Student_Name | Department_Name

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1 x student

```
1 • SELECT
2     Student.student_name AS Student_Name,
3     Department.department_name AS Department_Name
4 FROM Student
5 INNER JOIN Department
6 ON Student.department_id = Department.department_id;
```

Limit to 1000 rows

Result Grid

Student_Name	Department_Name
Aman	Computer Science
Rahul	Computer Science
Priya	Computer Science
Neha	Computer Science
Arjun	Computer Science
Karan	Computer Science
Simran	Information Technology
Riya	Information Technology
Pooja	Information Technology
Vikas	Electronics
Ankt	Electronics

Administration Schemas

Information

No object selected

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 15	19:52:46	SELECT Student.student_name, Department.department_name FR...	11 row(s) returned	0.000 sec / 0.000 sec
✓ 16	19:54:29	SELECT Student.student_name, Department.location FROM Stude...	11 row(s) returned	0.000 sec / 0.000 sec
✓ 17	19:55:09	SELECT Course.course_id, Course.course_name, Department.d...	7 row(s) returned	0.000 sec / 0.000 sec
✓ 18	19:55:40	SELECT Student.student_name, Course.course_name, Enrollme...	18 row(s) returned	0.000 sec / 0.000 sec
✓ 19	19:56:19	SELECT DISTINCT Student.student_name FROM Student INNER JOI...	11 row(s) returned	0.015 sec / 0.000 sec
✓ 20	19:57:37	SELECT Student.student_name AS Student_Name, Department.de...	11 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Read Only Context Help Snippets

Q7. Find the details of students belonging to the "Computer Science" department.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

SCHEMAS

- college
- college_management
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1 x student

Limit to 1000 rows

```
1 SELECT
2     Execute the selected portion of the script or everything, if there is no selection
3
4     Student.student_name,
5     Student.age,
6     Department.department_name
7 FROM Student
8 INNER JOIN Department
9 ON Student.department_id = Department.department_id
10 WHERE Department.department_name = 'Computer Science';
```

Result Grid

student_id	student_name	age	department_name
101	Aman	19	Computer Science
102	Rahul	20	Computer Science
103	Priya	19	Computer Science
104	Neha	21	Computer Science
105	Arjun	20	Computer Science
106	Karan	19	Computer Science

Administration Schemas

Information

No object selected

Result 9 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 16	19:54:29	SELECT Student.student_name, Department.location FROM Stude...	11 row(s) returned	0.000 sec / 0.000 sec
✓ 17	19:55:09	SELECT Course.course_id, Course.course_name, Department.d...	7 row(s) returned	0.000 sec / 0.000 sec
✓ 18	19:55:40	SELECT Student.student_name, Course.course_name, Enrolme...	18 row(s) returned	0.000 sec / 0.000 sec
✓ 19	19:56:19	SELECT DISTINCT Student.student_name FROM Student INNER JOI...	11 row(s) returned	0.015 sec / 0.000 sec
✓ 20	19:57:37	SELECT Student.student_name AS Student_Name, Department.de...	11 row(s) returned	0.000 sec / 0.000 sec
✓ 21	19:57:54	SELECT Student.student_id, Student.student_name, Student.age...	6 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Q8. Display: Student Name Course Name Marks for all enrolled students.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1 x student

Limit to 1000 rows

```
1 SELECT
2     Execute the selected portion of the script or everything, if there is no selection
3     Course.course_name AS Course_Name,
4     Enrollment.marks AS Marks
5 FROM Student
6 INNER JOIN Enrollment
7     ON Student.student_id = Enrollment.student_id
8 INNER JOIN Course
9     ON Enrollment.course_id = Course.course_id;
```

Result Grid

Student_Name	Course_Name	Marks
Aman	Database Management Systems	85
Rahul	Database Management Systems	92
Priya	Database Management Systems	76
Neha	Database Management Systems	65
Karan	Database Management Systems	55
Aman	Data Structures	78
Rahul	Data Structures	88
Arjun	Data Structures	95
Aman	Operating Systems	91
Priya	Operating Systems	84
Arjun	Operating Systems	89
Neha	Web Development	72
Simran	Web Development	81
Riya	Web Development	68

Administration Schemas

Information

No object selected

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 17	19:55:09	SELECT Course.course_id, Course.course_name, Department.d...	7 row(s) returned	0.000 sec / 0.000 sec
✓ 18	19:55:40	SELECT Student.student_name, Course.course_name, Enrollme...	18 row(s) returned	0.000 sec / 0.000 sec
✓ 19	19:56:19	SELECT DISTINCT Student.student_name FROM Student INNER JOI...	11 row(s) returned	0.015 sec / 0.000 sec
✓ 20	19:57:37	SELECT Student.student_name AS Student_Name, Department.de...	11 row(s) returned	0.000 sec / 0.000 sec
✓ 21	19:57:54	SELECT Student.student_id, Student.student_name, Student.age...	6 row(s) returned	0.000 sec / 0.000 sec
✓ 22	19:58:07	SELECT Student.student_name AS Student_Name, Course.course_...	18 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Read Only Context Help Snippets

Q9. Display department-wise student details using INNER JOIN.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1 x student

```
1 SELECT
2   Department.department_name AS Department_Name,
3   Student.student_id,
4   Student.student_name,
5   Student.age
6 FROM Department
7 INNER JOIN Student
8 ON Department.department_id = Student.department_id
9 ORDER BY Department.department_name;
```

Result Grid

Department_Name	student_id	student_name	age
Computer Science	101	Aman	19
Computer Science	102	Rahul	20
Computer Science	103	Priya	19
Computer Science	104	Neha	21
Computer Science	105	Arjun	20
Computer Science	106	Karan	19
Electronics	109	Vikas	21
Electronics	110	Ankit	20
Information Technology	107	Simran	20
Information Technology	108	Riya	19
Information Technology	113	Pooja	21

Administration Schemas

Information

No object selected

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Form Editor

Field Types

Query Stats

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 18	19:55:40	SELECT	Student.student_name, Course.course_name, Enrollme...	18 row(s) returned 0.000 sec / 0.000 sec
✓ 19	19:56:19	SELECT DISTINCT	Student.student_name FROM Student INNER JOI...	11 row(s) returned 0.015 sec / 0.000 sec
✓ 20	19:57:37	SELECT	Student.student_name AS Student_Name, Department.de...	11 row(s) returned 0.000 sec / 0.000 sec
✓ 21	19:57:54	SELECT	Student.student_id, Student.student_name, Student.age...	6 row(s) returned 0.000 sec / 0.000 sec
✓ 22	19:58:07	SELECT	Student.student_name AS Student_Name, Course.course_...	18 row(s) returned 0.000 sec / 0.000 sec
✓ 23	19:58:21	SELECT	Department.department_name AS Department_Name, Stu...	11 row(s) returned 0.016 sec / 0.000 sec

Object Info Session

LEFT JOIN Questions

Q10. Display all students along with their department names, including students who do not belong to any department.

The screenshot shows the MySQL Workbench interface. The 'Query 1' tab is active, displaying a SQL query that performs a LEFT JOIN between the 'Student' and 'Department' tables. The query is as follows:

```
1 SELECT
2     Execute the selected portion of the script or everything, if there is no selection
3     Student.student_name,
4     Department.department_name
5 FROM Student
6 LEFT JOIN Department
7 ON Student.department_id = Department.department_id;
```

The 'Result Grid' shows the output of the query, displaying 13 rows. The columns are 'student_id', 'student_name', and 'department_name'. The first 6 rows show students from the 'Computer Science' department, the next 3 rows show students from the 'Information Technology' department, and the last 4 rows show students from the 'Electronics' department. The 'department_name' column is NULL for students 111, 112, and 113.

student_id	student_name	department_name
101	Aman	Computer Science
102	Rahul	Computer Science
103	Priya	Computer Science
104	Neha	Computer Science
105	Arjun	Computer Science
106	Karan	Computer Science
107	Simran	Information Technology
108	Riya	Information Technology
109	Vikas	Electronics
110	Ankit	Electronics
111	Rohit	NULL
112	Sneha	NULL
113	Pooja	Information Technology

The 'Output' tab shows the execution log, indicating that the query was executed successfully and returned 13 rows. The log also shows the duration of the query execution.

Q11. Find students who are not enrolled in any course.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'college_management' selected. The main editor shows a SQL query: `SELECT Student.student_name FROM Student LEFT JOIN Enrollment ON Student.student_id = Enrollment.student_id WHERE Enrollment.enrollment_id IS NULL;`. The 'Result Grid' below the query shows two rows: student_id 111 (Rohit) and student_id 112 (Sneha). The bottom panel shows the 'Action Output' table with 6 rows of log messages.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

college

college_management

Tables

Views

Stored Procedures

Functions

sakila

sys

world

Query 1 student

Limit to 1000 rows

```
1 SELECT
2 Student.student_name
3 FROM Student
4 LEFT JOIN Enrollment
5 ON Student.student_id = Enrollment.student_id
6 WHERE Enrollment.enrollment_id IS NULL;
```

Result Grid

student_id	student_name
111	Rohit
112	Sneha

Administration Schemas

Information

No object selected

Result 13

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
20	19:57:37	SELECT	Student.student_name AS Student_Name, Department.de... 11 row(s) returned	0.000 sec / 0.000 sec
21	19:57:54	SELECT	Student.student_id, Student.student_name, Student.age... 6 row(s) returned	0.000 sec / 0.000 sec
22	19:58:07	SELECT	Student.student_name AS Student_Name, Course.course_... 18 row(s) returned	0.000 sec / 0.000 sec
23	19:58:21	SELECT	Department.department_name AS Department_Name, Stu... 11 row(s) returned	0.016 sec / 0.000 sec
24	19:58:36	SELECT	Student.student_id, Student.student_name, Department... 13 row(s) returned	0.016 sec / 0.000 sec
25	19:58:50	SELECT	Student.student_id, Student.student_name FROM Student... 2 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Q12. Display all departments and the students belonging to them, including departments with no students.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

college

college_management

Tables

Views

Stored Procedures

Functions

sakila

sys

world

Query 1 x student

Limit to 1000 rows

```
1 • SELECT
2   Department.department_name,
3   Student.student_name
4 FROM Department
5 LEFT JOIN Student
6 ON Department.department_id = Student.department_id
7 ORDER BY Department.department_name;
```

Result Grid

department_name	student_name
Civil	NULL
Computer Science	Aman
Computer Science	Rahul
Computer Science	Priya
Computer Science	Neha
Computer Science	Arjun
Computer Science	Karan
Electronics	Vikas
Electronics	Ankit
Information Technology	Simran
Information Technology	Riya
Information Technology	Pooja
Mechanical	NULL

Result 14 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
21	19:57:54	SELECT	Student.student_id, Student.student_name, Student.age... 6 row(s) returned	0.000 sec / 0.000 sec
22	19:58:07	SELECT	Student.student_name AS Student_Name, Course.course_... 18 row(s) returned	0.000 sec / 0.000 sec
23	19:58:21	SELECT	Department.department_name AS Department_Name, Stu... 11 row(s) returned	0.016 sec / 0.000 sec
24	19:58:36	SELECT	Student.student_id, Student.student_name, Department.... 13 row(s) returned	0.016 sec / 0.000 sec
25	19:58:50	SELECT	Student.student_id, Student.student_name FROM Student... 2 row(s) returned	0.000 sec / 0.000 sec
26	19:59:02	SELECT	Department.department_name, Student.student_name FR... 13 row(s) returned	0.000 sec / 0.000 sec

No object selected

Object Info Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Q13. Display all courses along with enrolled students, including courses having no enrollment.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Query 1 x student

```
1 • SELECT
2     Course.course_id,
3     Course.course_name,
4     Student.student_name
5 FROM Course
6 LEFT JOIN Enrollment
7     ON Course.course_id = Enrollment.course_id
8 LEFT JOIN Student
9     ON Enrollment.student_id = Student.student_id
```

Result Grid

course_id	course_name	student_name
201	Database Management Systems	Aman
201	Database Management Systems	Rahul
201	Database Management Systems	Priya
201	Database Management Systems	Neha
201	Database Management Systems	Karan
202	Data Structures	Aman
202	Data Structures	Rahul
202	Data Structures	Arjun
203	Operating Systems	Aman
203	Operating Systems	Priya
203	Operating Systems	Arjun
204	Web Development	Neha
204	Web Development	Simran
204	Web Development	Riya

Result 15 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 22	19:58:07	SELECT	Student.student_name AS Student_Name, Course.course_... 18 row(s) returned	0.000 sec / 0.000 sec
✓ 23	19:58:21	SELECT	Department.department_name AS Department_Name, Stu... 11 row(s) returned	0.016 sec / 0.000 sec
✓ 24	19:58:36	SELECT	Student.student_id, Student.student_name, Department... 13 row(s) returned	0.016 sec / 0.000 sec
✓ 25	19:58:50	SELECT	Student.student_id, Student.student_name FROM Student... 2 row(s) returned	0.000 sec / 0.000 sec
✓ 26	19:59:02	SELECT	Department.department_name, Student.student_name FR... 13 row(s) returned	0.000 sec / 0.000 sec
✓ 27	19:59:15	SELECT	Course.course_id, Course.course_name, Student.studen... 19 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Navigation: Schemas, Filter objects, college, college_management, Tables, Views, Stored Procedures, Functions, sakila, sys, world

SQLAdditions: Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Tools: Result Grid, Form Editor, Field Types, Query Stats

RIGHT JOIN Questions

Q14. Display all departments and their students using RIGHT JOIN.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: SCHEMAS

Filter objects

- college
- college_management
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1 x student

```
1 • SELECT
2   -- Exclude the selected portion of the script or everything, if there is no selection
3   Student.student_name
4 FROM Student
5 RIGHT JOIN Department
6 ON Student.department_id = Department.department_id;
```

Result Grid

department_name	student_name
Computer Science	Aman
Computer Science	Rahul
Computer Science	Priya
Computer Science	Neha
Computer Science	Arjun
Computer Science	Karan
Information Technology	Simran
Information Technology	Riya
Information Technology	Pooja
Electronics	Vikas
Electronics	Ankit
Mechanical	NULL
Civil	NULL

Result 16 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
23	19:58:21	SELECT	Department.department_name AS Department_Name, Stu...	11 row(s) returned 0.016 sec / 0.000 sec
24	19:58:36	SELECT	Student.student_id, Student.student_name, Department....	13 row(s) returned 0.016 sec / 0.000 sec
25	19:58:50	SELECT	Student.student_id, Student.student_name FROM Student...	2 row(s) returned 0.000 sec / 0.000 sec
26	19:59:02	SELECT	Department.department_name, Student.student_name FR...	13 row(s) returned 0.000 sec / 0.000 sec
27	19:59:15	SELECT	Course.course_id, Course.course_name, Student.studen...	19 row(s) returned 0.000 sec / 0.000 sec
28	19:59:29	SELECT	Department.department_name, Student.student_name FR...	13 row(s) returned 0.000 sec / 0.000 sec

Object Info Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Q15. Display all courses and their enrolled students using RIGHT JOIN.

The screenshot shows the MySQL Workbench interface with a query window titled 'Query 1' containing the following SQL code:

```
1 • SELECT
2   Exclude the selected portion of the script or everything, if there is no selection
3   Student.student_name
4 FROM Enrollment
5 RIGHT JOIN Course
6 ON Enrollment.course_id = Course.course_id
7 LEFT JOIN Student
8 ON Enrollment.student_id = Student.student_id;
```

The 'Result Grid' displays the following data:

course_name	student_name
Database Management Systems	Aman
Database Management Systems	Rahul
Database Management Systems	Priya
Database Management Systems	Neha
Database Management Systems	Karan
Data Structures	Aman
Data Structures	Rahul
Data Structures	Arjun
Operating Systems	Aman
Operating Systems	Priya
Operating Systems	Arjun
Web Development	Neha
Web Development	Simran
Web Development	Riya

The 'Output' window shows the execution log:

#	Time	Action	Message	Duration / Fetch
✓ 24	19:58:36	SELECT Student.student_id, Student.student_name, Department....	13 row(s) returned	0.016 sec / 0.000 sec
✓ 25	19:58:50	SELECT Student.student_id, Student.student_name FROM Student...	2 row(s) returned	0.000 sec / 0.000 sec
✓ 26	19:59:02	SELECT Department.department_name, Student.student_name FR...	13 row(s) returned	0.000 sec / 0.000 sec
✓ 27	19:59:15	SELECT Course.course_id, Course.course_name, Student.studen...	19 row(s) returned	0.000 sec / 0.000 sec
✓ 28	19:59:29	SELECT Department.department_name, Student.student_name FR...	13 row(s) returned	0.000 sec / 0.000 sec
✓ 29	19:59:42	SELECT Course.course_name, Student.student_name FROM Enroll...	19 row(s) returned	0.000 sec / 0.000 sec

Multiple JOIN Questions

Q16. Display complete student course information: Student Name Department Name Course Name Marks

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- college
- college_management
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1 student

```
1 • SELECT
2     Student.student_name AS Student_Name,
3     Department.department_name AS Department_Name,
4     Course.course_name AS Course_Name,
5     Enrollment.marks AS Marks
6 FROM Student
7 INNER JOIN Department
8     ON Student.department_id = Department.department_id
9 INNER JOIN Enrollment
```

Result Grid

Student_Name	Department_Name	Course_Name	Marks
Aman	Computer Science	Database Management Systems	85
Aman	Computer Science	Data Structures	78
Aman	Computer Science	Operating Systems	91
Rahul	Computer Science	Database Management Systems	92
Rahul	Computer Science	Data Structures	88
Priya	Computer Science	Database Management Systems	76
Priya	Computer Science	Operating Systems	84
Neha	Computer Science	Database Management Systems	65
Neha	Computer Science	Web Development	72
Arjun	Computer Science	Data Structures	95
Arjun	Computer Science	Operating Systems	89
Karan	Computer Science	Database Management Systems	55
Simran	Information Tech...	Web Development	81
Simran	Information Tech...	Computer Networks	74

Result 18

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 25	19:58:50	SELECT Student.student_id, Student.student_name FROM Student...	2 row(s) returned	0.000 sec / 0.000 sec
✓ 26	19:59:02	SELECT Department.department_name, Student.student_name FR...	13 row(s) returned	0.000 sec / 0.000 sec
✓ 27	19:59:15	SELECT Course.course_id, Course.course_name, Student.studen...	19 row(s) returned	0.000 sec / 0.000 sec
✓ 28	19:59:29	SELECT Department.department_name, Student.student_name FR...	13 row(s) returned	0.000 sec / 0.000 sec
✓ 29	19:59:42	SELECT Course.course_name, Student.student_name FROM Enroll...	19 row(s) returned	0.000 sec / 0.000 sec
✓ 30	19:59:55	SELECT Student.student_name AS Student_Name, Department.de...	18 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Read Only Context Help Snippets

Q17. Display: Student Name Department Location Course Name using multiple JOIN operations.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

college

college_management

Tables

Views

Stored Procedures

Functions

sakila

sys

world

Query 1 x student

Limit to 1000 rows

```
1 • SELECT
2     Student.student_name AS Student_Name,
3     Department.location AS Department_Location,
4     Course.course_name AS Course_Name
5 FROM Student
6 INNER JOIN Department
7     ON Student.department_id = Department.department_id
8 INNER JOIN Enrollment
9     ON Student.student_id = Enrollment.student_id
```

Result Grid

	Student_Name	Department_Location	Course_Name
▶	Aman	Delhi	Database Management Systems
	Aman	Delhi	Data Structures
	Aman	Delhi	Operating Systems
	Rahul	Delhi	Database Management Systems
	Rahul	Delhi	Data Structures
	Priya	Delhi	Database Management Systems
	Priya	Delhi	Operating Systems
	Neha	Delhi	Database Management Systems
	Neha	Delhi	Web Development
	Arjun	Delhi	Data Structures
	Arjun	Delhi	Operating Systems
	Karan	Delhi	Database Management Systems
	Simran	Noida	Web Development
	Simran	Noida	Computer Networks

Administration Schemas

Information

No object selected

Result 21 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 28	19:59:29	SELECT Department.department_name, Student.student_name FR...	13 row(s) returned	0.000 sec / 0.000 sec
✓ 29	19:59:42	SELECT Course.course_name, Student.student_name FROM Enroll...	19 row(s) returned	0.000 sec / 0.000 sec
✓ 30	19:59:55	SELECT Student.student_name AS Student_Name, Department.de...	18 row(s) returned	0.000 sec / 0.000 sec
✓ 31	20:00:11	SELECT Student.student_name AS Student_Name, Department.loc...	18 row(s) returned	0.000 sec / 0.000 sec
✓ 32	20:00:18	SELECT Student.student_name AS Student_Name, Department.de...	18 row(s) returned	0.000 sec / 0.000 sec
✓ 33	20:00:21	SELECT Student.student_name AS Student_Name, Department.loc...	18 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Q18. Find the average marks obtained by students in each course along with the course name.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'college_management' selected. The main editor shows a SQL query for 'Query 1' in the 'student' database. The query is as follows:

```
1 SELECT
2   Course.course_name,
3   AVG(Enrollment.marks) AS Average_Marks
4 FROM Course
5 INNER JOIN Enrollment
6 ON Course.course_id = Enrollment.course_id
7 GROUP BY Course.course_id, Course.course_name;
```

Below the query editor, the 'Result Grid' shows the output of the query:

course_name	Average_Marks
Database Management Systems	74.6000
Data Structures	87.0000
Operating Systems	88.0000
Web Development	73.6667
Computer Networks	78.0000
Digital Electronics	62.5000

The bottom panel shows the 'Output' tab with a table of actions and their results:

#	Time	Action	Message	Duration / Fetch
29	19:59:42	SELECT Course.course_name, Student.student_name FROM Enroll...	19 row(s) returned	0.000 sec / 0.000 sec
30	19:59:55	SELECT Student.student_name AS Student_Name, Department.de...	18 row(s) returned	0.000 sec / 0.000 sec
31	20:00:11	SELECT Student.student_name AS Student_Name, Department.loc...	18 row(s) returned	0.000 sec / 0.000 sec
32	20:00:18	SELECT Student.student_name AS Student_Name, Department.de...	18 row(s) returned	0.000 sec / 0.000 sec
33	20:00:21	SELECT Student.student_name AS Student_Name, Department.loc...	18 row(s) returned	0.000 sec / 0.000 sec
34	20:00:36	SELECT Course.course_name, AVG(Enrollment.marks) AS Average...	6 row(s) returned	0.016 sec / 0.000 sec

Q19. Display department-wise: Department Name Number of Students using JOIN and GROUP BY.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Query 1 x student

Limit to 1000 rows

SQL

SCHEMAS

Filter objects

college

college_management

Tables

Views

Stored Procedures

Functions

sakila

sys

world

1 • SELECT

2

3

4

5

6

7

Execute the selected portion of the script or everything, if there is no selection

COUNT(Student.student_id) AS Number_of_Students

FROM Department

LEFT JOIN Student

ON Department.department_id = Student.department_id

GROUP BY Department.department_id, Department.department_name;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Result Grid

Form Editor

Field Types

Query Stats

Administration Schemas

Information

No object selected

Result 23 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 30	19:59:55	SELECT Student.student_name AS Student_Name, Department.de...	18 row(s) returned	0.000 sec / 0.000 sec
✓ 31	20:00:11	SELECT Student.student_name AS Student_Name, Department.loc...	18 row(s) returned	0.000 sec / 0.000 sec
✓ 32	20:00:18	SELECT Student.student_name AS Student_Name, Department.de...	18 row(s) returned	0.000 sec / 0.000 sec
✓ 33	20:00:21	SELECT Student.student_name AS Student_Name, Department.loc...	18 row(s) returned	0.000 sec / 0.000 sec
✓ 34	20:00:36	SELECT Course.course_name, AVG(Enrollment.marks) AS Average...	6 row(s) returned	0.016 sec / 0.000 sec
✓ 35	20:00:47	SELECT Department.department_name, COUNT(Student.student_...	5 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Q20. Display: Department Name Course Name Number of Students Enrolled

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

- college
- college_management
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1 x student

```
1 • SELECT
2     Department.department_name AS Department_Name,
3     Course.course_name AS Course_Name,
4     COUNT(Enrollment.student_id) AS Number_of_Students_Enrolled
5 FROM Department
6 INNER JOIN Course
7     ON Department.department_id = Course.department_id
8 LEFT JOIN Enrollment
9     ON Course.course_id = Enrollment.course_id
```

Result Grid

	Department_Name	Course_Name	Number_of_Students_Enrolled
▶	Computer Science	Database Management Systems	5
	Computer Science	Data Structures	3
	Computer Science	Operating Systems	3
	Information Technology	Web Development	3
	Information Technology	Computer Networks	2
	Electronics	Digital Electronics	2
	Mechanical	Thermodynamics	0

Administration Schemas

Information

No object selected

Result 24 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 31	20:00:11	SELECT	Student.student_name AS Student_Name, Department.loc... 18 row(s) returned	0.000 sec / 0.000 sec
✓ 32	20:00:18	SELECT	Student.student_name AS Student_Name, Department.de... 18 row(s) returned	0.000 sec / 0.000 sec
✓ 33	20:00:21	SELECT	Student.student_name AS Student_Name, Department.loc... 18 row(s) returned	0.000 sec / 0.000 sec
✓ 34	20:00:36	SELECT	Course.course_name, AVG(Enrollment.marks) AS Average... 6 row(s) returned	0.016 sec / 0.000 sec
✓ 35	20:00:47	SELECT	Department.department_name, COUNT(Student.student_i... 5 row(s) returned	0.016 sec / 0.000 sec
✓ 36	20:01:00	SELECT	Department.department_name AS Department_Name, Cou... 7 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Advanced JOIN Questions

Q21. Find the student who scored the highest marks in each course.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'college_management' selected. The main editor shows 'Query 1' with the following SQL:

```
1 • SELECT
2     Course.course_name,
3     Student.student_name,
4     Enrollment.marks
5 FROM Enrollment
6 INNER JOIN Student
7     ON Enrollment.student_id = Student.student_id
8 INNER JOIN Course
9     ON Enrollment.course_id = Course.course_id
```

Below the query editor, the 'Result Grid' shows the following data:

course_name	student_name	marks
Database Management Systems	Rahul	92
Data Structures	Arjun	95
Operating Systems	Aman	91
Web Development	Simran	81
Computer Networks	Pooja	82
Digital Electronics	Vikas	90

The bottom panel shows the 'Output' tab with 'Action Output' selected, displaying a log of SQL actions and their execution details.

#	Time	Action	Message	Duration / Fetch
32	20:00:18	SELECT	Student.student_name AS Student_Name, Department.de...	18 row(s) returned 0.000 sec / 0.000 sec
33	20:00:21	SELECT	Student.student_name AS Student_Name, Department.loc...	18 row(s) returned 0.000 sec / 0.000 sec
34	20:00:36	SELECT	Course.course_name, AVG(Enrollment.marks) AS Average...	6 row(s) returned 0.016 sec / 0.000 sec
35	20:00:47	SELECT	Department.department_name, COUNT(Student.student_j...	5 row(s) returned 0.016 sec / 0.000 sec
36	20:01:00	SELECT	Department.department_name AS Department_Name, Cou...	7 row(s) returned 0.000 sec / 0.000 sec
37	20:01:13	SELECT	Course.course_name, Student.student_name, Enrollmen...	6 row(s) returned 0.000 sec / 0.000 sec

Q22. Display departments where no student is registered.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'college_management' selected. The main editor shows a SQL query for departments with no students. The 'Result Grid' shows two rows: Mechanical and Civil. The 'Output' pane at the bottom shows a list of actions and their results.

Query 1: student

```
1 SELECT
2   Department.department_id,
3   Department.department_name
4 FROM Department
5 LEFT JOIN Student
6 ON Department.department_id = Student.department_id
7 WHERE Student.student_id IS NULL;
```

Result Grid

department_id	department_name
4	Mechanical
5	Civil

Output

#	Time	Action	Message	Duration / Fetch
33	20:00:21	SELECT	Student.student_name AS Student_Name, Department.loc...	18 row(s) returned 0.000 sec / 0.000 sec
34	20:00:36	SELECT	Course.course_name, AVG(Enrollment.marks) AS Average...	6 row(s) returned 0.016 sec / 0.000 sec
35	20:00:47	SELECT	Department.department_name, COUNT(Student.student_id)	5 row(s) returned 0.016 sec / 0.000 sec
36	20:01:00	SELECT	Department.department_name AS Department_Name, Cou...	7 row(s) returned 0.000 sec / 0.000 sec
37	20:01:13	SELECT	Course.course_name, Student.student_name, Enrollmen...	6 row(s) returned 0.000 sec / 0.000 sec
38	20:01:24	SELECT	Department.department_id, Department.department_name ...	2 row(s) returned 0.000 sec / 0.000 sec

Q23. Display courses in which no student has enrolled.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'college_management' selected. The central editor shows a SQL query: `SELECT Course.course_id, Course.course_name FROM Course LEFT JOIN Enrollment ON Course.course_id = Enrollment.course_id WHERE Enrollment.enrollment_id IS NULL;`. Below the query, the 'Result Grid' shows one result: course_id 207, course_name Thermodynamics. The bottom panel shows the 'Output' tab with a table of actions and their results.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

college

college_management

Tables

Views

Stored Procedures

Functions

sakila

sys

world

Query 1 student

Limit to 1000 rows

1 SELECT

2 Course.course_id,

3 Course.course_name

4 FROM Course

5 LEFT JOIN Enrollment

6 ON Course.course_id = Enrollment.course_id

7 WHERE Enrollment.enrollment_id IS NULL;

Result Grid

course_id course_name

207 Thermodynamics

Administration Schemas

Information

No object selected

Result 27

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
34	20:00:36	SELECT Course.course_name, AVG(Enrollment.marks) AS Average...	6 row(s) returned	0.016 sec / 0.000 sec
35	20:00:47	SELECT Department.department_name, COUNT(Student.student_id)...	5 row(s) returned	0.016 sec / 0.000 sec
36	20:01:00	SELECT Department.department_name AS Department_Name, Cou...	7 row(s) returned	0.000 sec / 0.000 sec
37	20:01:13	SELECT Course.course_name, Student.student_name, Enrollmen...	6 row(s) returned	0.000 sec / 0.000 sec
38	20:01:24	SELECT Department.department_id, Department.department_name ...	2 row(s) returned	0.000 sec / 0.000 sec
39	20:01:35	SELECT Course.course_id, Course.course_name FROM Course LE...	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Q24. Find the total number of students in each department.

The screenshot shows the MySQL Workbench interface. The 'Navigator' pane on the left displays the database schema, including the 'college_management' database. The 'Query Editor' pane in the center contains the following SQL query:

```
1 • SELECT
2     Department.department_name,
3     COUNT(Student.student_id) AS Total_Students
4 FROM Department
5 LEFT JOIN Student
6 ON Department.department_id = Student.department_id
7 GROUP BY
8     Department.department_id,
9     Department.department_name;
```

The 'Result Grid' pane below the query editor shows the results of the query:

department_name	Total_Students
Computer Science	6
Information Technology	3
Electronics	2
Mechanical	0
Civil	0

The 'Output' pane at the bottom shows the execution log, including the query text and the number of rows returned for each statement.

#	Time	Action	Message	Duration / Fetch
35	20:00:47	SELECT	Department.department_name, COUNT(Student.student_id)...	5 row(s) returned 0.016 sec / 0.000 sec
36	20:01:00	SELECT	Department.department_name AS Department_Name, Cou...	7 row(s) returned 0.000 sec / 0.000 sec
37	20:01:13	SELECT	Course.course_name, Student.student_name, Enrollmen...	6 row(s) returned 0.000 sec / 0.000 sec
38	20:01:24	SELECT	Department.department_id, Department.department_name ...	2 row(s) returned 0.000 sec / 0.000 sec
39	20:01:35	SELECT	Course.course_id, Course.course_name FROM Course LE...	1 row(s) returned 0.000 sec / 0.000 sec
40	20:01:47	SELECT	Department.department_name, COUNT(Student.student_id)...	5 row(s) returned 0.000 sec / 0.000 sec

Q25. Display the complete report: Student ID Student Name Department Course Marks Result Status
 where: • Marks $\geq 40 \rightarrow$ "Pass" • Marks $< 40 \rightarrow$ "Fail"

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: SCHEMAS

- college
 - college_management
 - Tables
 - Views
 - Stored Procedures
 - Functions
 - sakila
 - sys
 - world

Query 1 x student

```

6      Enrollment.marks AS Marks,
7      CASE
8          WHEN Enrollment.marks >= 40 THEN 'Pass'
9          ELSE 'Fail'
10     END AS Result_Status
11 FROM Student
12 INNER JOIN Department
13     ON Student.department_id = Department.department_id
14 INNER JOIN Enrollment
15     ON Student.student_id = Enrollment.student_id
16 INNER JOIN Course
17     ON Enrollment.course_id = Course.course_id;
    
```

Result Grid

Student_ID	Student_Name	Department	Course	Marks	Result_Status
101	Aman	Computer Science	Database Management Systems	85	Pass
101	Aman	Computer Science	Data Structures	78	Pass
101	Aman	Computer Science	Operating Systems	91	Pass
102	Rahul	Computer Science	Database Management Systems	92	Pass
102	Rahul	Computer Science	Data Structures	88	Pass
103	Priya	Computer Science	Database Management Systems	76	Pass
103	Priya	Computer Science	Operating Systems	84	Pass
104	Neha	Computer Science	Database Management Systems	65	Pass
104	Neha	Computer Science	Web Development	72	Pass
105	Arun	Computer Science	Data Structures	65	Pass

Result 29 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
36	20:01:00	SELECT	Department.department_name AS Department_Name, Cou...	7 row(s) returned 0.000 sec / 0.000 sec
37	20:01:13	SELECT	Course.course_name, Student.student_name, Enrollmen...	6 row(s) returned 0.000 sec / 0.000 sec
38	20:01:24	SELECT	Department.department_id, Department.department_name ...	2 row(s) returned 0.000 sec / 0.000 sec
39	20:01:35	SELECT	Course.course_id, Course.course_name FROM Course LE...	1 row(s) returned 0.000 sec / 0.000 sec
40	20:01:47	SELECT	Department.department_name, COUNT(Student.student_i...	5 row(s) returned 0.000 sec / 0.000 sec
41	20:02:09	SELECT	Student.student_id AS Student_ID, Student.student_name...	18 row(s) returned 0.016 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Read Only Context Help Snippets

Self JOIN

Practice Assume another table: Table: Employee Column Data Type emp_id INT emp_name VARCHAR(50) manager_id INT

Q26. Display employee name with their manager name using SELF JOIN.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'college_management' selected. The central editor shows a SQL query for 'Query 1' with the following code:

```
1 SELECT
2     E.emp_name AS Employee_Name,
3     M.emp_name AS Manager_Name
4 FROM Employee AS E
5 LEFT JOIN Employee AS M
6 ON E.manager_id = M.emp_id;
```

Below the query editor, the 'Result Grid' shows the output of the query:

Employee_Name	Manager_Name
Raj	NULL
Amit	Raj
Priya	Raj
Neha	Amit
Karan	Amit

The bottom panel shows the 'Action Output' tab with a log of database actions:

#	Time	Action	Message	Duration / Fetch
39	20:01:35	SELECT	Course.course_id, Course.course_name FROM Course LE...	1 row(s) returned 0.000 sec / 0.000 sec
40	20:01:47	SELECT	Department.department_name, COUNT(Student.student_j...	5 row(s) returned 0.000 sec / 0.000 sec
41	20:02:09	SELECT	Student.student_id AS Student_ID, Student.student_name...	18 row(s) returned 0.016 sec / 0.000 sec
42	20:02:21	CREATE TABLE	Employee (emp_id INT PRIMARY KEY, emp_nam...	0 row(s) affected 0.031 sec
43	20:02:29	INSERT INTO	Employee VALUES (1, 'Raj', NULL), (2, 'Amit', 1), (3, 'Priya'...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 0.016 sec
44	20:02:40	SELECT	E.emp_name AS Employee_Name, M.emp_name AS Man...	5 row(s) returned 0.000 sec / 0.000 sec

Challenge Questions

Q27. Find departments having more than 5 students using JOIN and GROUP BY.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'college_management' selected. The main editor shows a SQL query for 'Query 1' named 'student'. The query is as follows:

```
1 • SELECT
2   COUNT(Student.student_id) AS Number_of_Students
3 FROM Department
4 INNER JOIN Student
5 ON Department.department_id = Student.department_id
6 GROUP BY
7   Department.department_id,
8   Department.department_name
9 HAVING COUNT(Student.student_id) > 5;
```

Below the query editor, the 'Result Grid' shows the results of the query:

department_name	Number_of_Students
Computer Science	6

The bottom panel shows the 'Output' tab with a table of actions and their results:

#	Time	Action	Message	Duration / Fetch
40	20:01:47	SELECT Department.department_name, COUNT(Student.student_id)...	5 row(s) returned	0.000 sec / 0.000 sec
41	20:02:09	SELECT Student.student_id AS Student_ID, Student.student_name...	18 row(s) returned	0.016 sec / 0.000 sec
42	20:02:21	CREATE TABLE Employee (emp_id INT PRIMARY KEY, emp_name...	0 row(s) affected	0.031 sec
43	20:02:29	INSERT INTO Employee VALUES (1, 'Raj', NULL), (2, 'Amit', 1), (3, 'Priya'...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
44	20:02:40	SELECT E.emp_name AS Employee_Name, M.emp_name AS Man...	5 row(s) returned	0.000 sec / 0.000 sec
45	20:02:55	SELECT Department.department_name, COUNT(Student.student_id)...	1 row(s) returned	0.015 sec / 0.000 sec

Q28. Find the second highest marks obtained by students using JOIN.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'college_management' selected. The main editor shows a SQL query in 'Query 1' titled 'student'.

```
1 SELECT
2     E1.marks AS Second_Highest_Marks
3 FROM Enrollment AS E1
4 LEFT JOIN Enrollment AS E2
5     ON E2.marks > E1.marks
6 GROUP BY E1.marks
7 HAVING COUNT(DISTINCT E2.marks) = 1;
```

The 'Result Grid' below the query shows a single row with the value '92' under the column 'Second_Highest_Marks'.

The bottom panel shows the 'Output' tab with a table of actions and messages:

#	Time	Action	Message	Duration / Fetch
41	20:02:09	SELECT Student.student_id AS Student_ID, Student.student_name...	18 row(s) returned	0.016 sec / 0.000 sec
42	20:02:21	CREATE TABLE Employee (emp_id INT PRIMARY KEY, emp_nam...	0 row(s) affected	0.031 sec
43	20:02:29	INSERT INTO Employee VALUES (1, 'Raj', NULL), (2, 'Amit', 1), (3, 'Priya'...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
44	20:02:40	SELECT E.emp_name AS Employee_Name, M.emp_name AS Man...	5 row(s) returned	0.000 sec / 0.000 sec
45	20:02:55	SELECT Department.department_name, COUNT(Student.student_id)	1 row(s) returned	0.015 sec / 0.000 sec
46	20:03:05	SELECT E1.marks AS Second_Highest_Marks FROM Enrollment AS ...	1 row(s) returned	0.015 sec / 0.000 sec

Q29. Display students who have enrolled in more than one course.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'college_management' selected. The main editor shows a SQL query for Query 1, which is a SELECT statement with an INNER JOIN and a HAVING clause. The query is as follows:

```
1 SELECT
2   Student.student_name,
3   COUNT(Enrollment.course_id) AS Number_of_Courses
4 FROM Student
5 INNER JOIN Enrollment
6 ON Student.student_id = Enrollment.student_id
7 GROUP BY
8   Student.student_id,
9   Student.student_name
10 HAVING COUNT(Enrollment.course_id) > 1;
```

Below the query editor, the 'Result Grid' shows the results of the query. The grid has three columns: 'student_id', 'student_name', and 'Number_of_Courses'. The results are as follows:

student_id	student_name	Number_of_Courses
101	Aman	3
102	Rahul	2
103	Priya	2
104	Neha	2
105	Arjun	2
107	Simran	2

At the bottom, the 'Output' tab shows a log of database actions. The log includes the following entries:

#	Time	Action	Message	Duration / Fetch
42	20:02:21	CREATE TABLE Employee (emp_id INT PRIMARY KEY, emp_nam...	0 row(s) affected	0.031 sec
43	20:02:29	INSERT INTO Employee VALUES (1, 'Raj', NULL), (2, 'Amit', 1), (3, 'Priya'...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
44	20:02:40	SELECT E.emp_name AS Employee_Name, M.emp_name AS Man...	5 row(s) returned	0.000 sec / 0.000 sec
45	20:02:55	SELECT Department.department_name, COUNT(Student.student_j...	1 row(s) returned	0.015 sec / 0.000 sec
46	20:03:05	SELECT E1.marks AS Second_Highest_Marks FROM Enrollment AS ...	1 row(s) returned	0.015 sec / 0.000 sec
47	20:03:15	SELECT Student.student_id, Student.student_name, COUNT(Enr...	6 row(s) returned	0.000 sec / 0.000 sec

Q30. Generate a complete college report showing: Department ↓ Students ↓ Courses ↓ Marks using appropriate JOIN operations.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

5      Enrollment.marks AS Marks
6  FR Execute the selected portion of the script or everything, if there is no selection
7  LEFT JOIN Student
8      ON Department.department_id = Student.department_id
9  LEFT JOIN Enrollment
10     ON Student.student_id = Enrollment.student_id
11 LEFT JOIN Course
12     ON Enrollment.course_id = Course.course_id
13 ORDER BY
14     Department.department_name,
15     Student.student_name,
16     Course.course_name;

```

The query results are displayed in the Result Grid, showing 34 rows. The columns are Department, Student, Course, and Marks. The data is as follows:

Department	Student	Course	Marks
Civil			
Computer Science	Aman	Data Structures	78
Computer Science	Aman	Database Management Systems	85
Computer Science	Aman	Operating Systems	91
Computer Science	Arjun	Data Structures	95
Computer Science	Arjun	Operating Systems	89
Computer Science	Karan	Database Management Systems	55
Computer Science	Neha	Database Management Systems	65
Computer Science	Neha	Web Development	72
Computer Science	Divya	Database Management Systems	76

The Output pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
43	20:02:29	INSERT INTO Employee VALUES (1, 'Raj', NULL), (2, 'Amit', 1), (3, 'Priya'...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
44	20:02:40	SELECT E.emp_name AS Employee_Name, M.emp_name AS Man...	5 row(s) returned	0.000 sec / 0.000 sec
45	20:02:55	SELECT Department.department_name, COUNT(Student.student_i...	1 row(s) returned	0.015 sec / 0.000 sec
46	20:03:05	SELECT E1.marks AS Second_Highest_Marks FROM Enrollment AS ...	1 row(s) returned	0.015 sec / 0.000 sec
47	20:03:15	SELECT Student.student_id, Student.student_name, COUNT(Enr...	6 row(s) returned	0.000 sec / 0.000 sec
48	20:03:37	SELECT Department.department_name AS Department, Student.stu...	20 row(s) returned	0.000 sec / 0.000 sec